MILITARY SPECIFICATION
MOUNT, TELESCOPE: M138

1. SCOPE

1.1 This specification covers performance requirements, quality assurance provisions, and packaging data relative to a telescope mount designed primarily to provide adjustments in elevation and azimuth for direct fire (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

sTANDARDS

Military

MIL-sTD-lo5	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-lo9	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

F8587500	Mount, Telescope: M138

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T05:30Z
Check the source to verify that this is the current version before use.

PACKAGING DATA SHEET

8587500 Packaging of Mount, Telescope: M138

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The mount shall be manufactured in accordance with Drawing F8587500 and drawings pertaining thereto.

3.2 General specification. - The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Part identification and marking
(g) Workmanship

3.3 Environmental.

3.3.1 Operating temperatures. - The mount shall meet the applicable requirements of 3.4.3 while exposed and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit (F) subsequent to which the mount shall meet the requirements of 3.3.2.

3.3.2 Vibration. - With the mount in the normal operating position, it shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration, the mount shall show no evidence of loose or damaged parts and the mount shall meet the requirements of 3.3.3 to 3.5 inclusive.

3.3.3 Sealing. - The mount shall meet the requirements of Procedure I of the rain test of MIL-F-13926.

3.4 Performance.

3.4.1 Orientation.- The mount shall be positioned as follows:

(a) The flat mounting surfaces of mount in a horizontal plane.
(b) The axis of the telescope socket parallel to the flat mounting surface and parallel to the plane of the centers of the two mounting bolt holes on the left side of the mount.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:30Z
Check the source to verify that this is the current version before use.

3.4.2 Locks.

(a) The telescope locking device shall lock and securely hold a 1.375 minus 0.001 inch diameter surface representative of the telescope (see Figure I).

(b) A torque of 4 pound-inches applied to the azimuth eccentric Plug 8587407 shall not change the azimuth alignment of the mount after the eccentric locking screw has been tightened with a torque of 15 ± 1 pound-inches.

(c) A torque of 5 pound inches applied to the elevation adjustment screw shall not change the elevation alignment of the mount.

3.4.3 Movement.

3.4.3.1 Deflection. - Rotation of the deflection (azimuth) eccentric Plug 8587407 shall move the adapter at least 12 mils to the right and 12 mils to the left from the position established in 3.4.1.

3.4.3.2 Elevation. - Rotation of the elevation adjustment screw shall move the telescope socket at least 25 degrees (444.45 mils) above and 25 degrees (444.45 mils) below the position established in 3.4.1.

3.4.4 Torque. - With boresight locks released, the running torque required to rotate the adjustment screw for elevation and deflection adjustment shall not exceed 5 pound-inches at standard ambient temperature (plus 60 degrees to 90 degrees F). The torque shall not exceed the values recorded at standard ambient temperature by more than 5 pound-inches when subjected to extreme temperature of 3.3.1. The rheostat shall be operable at extreme temperatures.

3.4.5 Instrument light. - The instrument light shall be checked for functional operation and illumination when the contacts of the mount are energized with 24 Vdc. Intensity of the lamp shall vary when the rheostat knob is actuated.

3.5 Electrical.

3.5.1 Insulation resistance.- Insulation resistance shall not be less than 10 megohms with a potential of 100 volts direct current applied.

3.5.2 Continuity.- The continuity of the wiring shall be checked for conformance with the applicable wiring and continuity drawing requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:30Z
Check the source to verify that this is the current version before use.

other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mounts and three (3) each of every component and subassembly covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that" it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL) when provided with the technical data package (TDP). Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables I and II. Examinations and tests for Packaging, Packing and Marking

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:30Z
Check the source to verify that this is the current version before use.

shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Vibration	3.3.2	4.6.2
102. Locks	3.4.2	4.6.5
103. Deflection	3.4.3.1	4.6.6
104. Elevation	3.4.3.2	4.6.7
105. Torque	3.4.4	4.6.8
106. Instrument Light	3.4.5	4.6.9
107. Insulation Resistance	3.5.1	4.6.10.1
108. Continuity	3.5.2	4.6.10.2

MINOR: NONE

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Environmental.- Three (3) mounts shall be selected at random from each 50 produced or from each month's production, whichever occurs first, for the following tests specified in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301. Operating temperature	3.3.1	4.6.1
302. Sealing	3.3.3	4.6.3

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:30Z
Check the source to verify that this is the current version before use.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified i the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL), when provided with the Technical Data Package (TDP).

4.5.2.2 contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and with the "component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment,

4.6 Test methods and procedures.

4.6.1 Operating temperatures.- The mounts shall be subjected to the temperatures specified in 3.3.1 utilizing the test facilities referenced in 4.5.2.2. The temperature shall be changed gradually to avoid thermal shock. When the mount has been thermally stabilized at each extreme temperature it shall meet the requirements of 3.4.3 and 3.4.4. Upon return to standard ambient temperature, the mount shall be tested in accordance with all remaining tests specified herein to determine compliance with respective requirements referenced in each test.

4.6.2 Vibration. - The mount shall be mounted on a vibration machine equipped with facilities to securely hold the mount while the machine is in motion. The mount shall be vibrated in accordance with 3.3.2 at the amplitude and for the period of time specified. Subsequent to vibration, a visual examination shall be made to determine evidence of physical failure or damage and the mount shall then meet all requirements as defined in 3.3.2.

4.6.3 Sealing. - This test shall be performed as specified in the rain test, Procedure I of MIL-F-13926. Subsequent to this test, the mount shall be examined by visual and tactile means for evidence of moisture or damage, then successfully tested for the performance requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:30Z
Check the source to verify that this is the current version before use.

4.6.4 Orientation.

(a) The mounting surfaces of the mount shall be positioned and secured on a mounting plate that is level within 1.0 mil. The axis through the telescope socket shall be parallel to the flat mounting surfaces within 3 minutes (0°3′).

(b) Secure the auxiliary telescope, held by bushing (Figure I), in the mount socket with telescope locking clamp. The reticle cross line pattern and target shall be aligned.

(c) The elevation locking lever shall be in the locked position and torque of 15 ± 1 pound-inches to the deflection locking screw shall be applied.

4.6.5 Locks. - The mount shall be positioned as specified in 4.6.4 with the auxiliary telescope aligned on the target. A torque of 4 pound-inches applied to the deflection eccentric plug shall not change the line of sight of the auxiliary telescope. A torque of 5 pound-inches applied to the elevation adjustment worm shall not change the line of sight of the auxiliary telescope.

4.6.6 Deflection.- The mount shall be in position as established in 4.6.4. Loosen the deflection locking screw. Rotate the deflection (azimuth) eccentric plug to move the auxiliary telescope to the right and then to the left. During this rotation view through the auxiliary telescope and ascertain if the movement, as measured by the target markings, meets the minimum requirements specified in 3.4.3.1. Return line of sight on target and lock deflection locking screw as specified in 4.6.4.

4.6.7 Elevation. - The mount shall be positioned as established in 4.6.4. Rotate the elevation locking lever clockwise to release the elevation mechanism. Rotate the elevation adjustment screw to move the auxiliary telescope up and then down. During this rotation, view through the auxiliary telescope and ascertain if the movement, as measured by the target markings, meets the minimum requirements specified in 3.4.3.2.

4.6.8 Torque. - With the mount stabilized at standard ambient temperature and the torque adapter attached to the deflection adjustment screw, the amount of torque required to rotate the screw must be within the specified tolerance and shall be recorded. This test is repeated at both extreme temperatures and the results obtained must not exceed the recorded values by more than the specified requirement. The entire test is repeated with the torque adapter attached to the elevation adjustment screw. While the mount is subjected to the extreme temperatures the toggle switch is actuated and must function.

4.6.9 Instrument light. - With a potential of 24 volts applied to the electrical contact of the mount the instrument light bulb shall be illuminated when the rheostat is actuated.

Source: https://assist.dla.mil -- Downloaded: 202640-22T05:30Z
Check the source to verify that this is the current version before use.

4.6.10 Electrical.

4.6.10.1 Insulation resistance.- The insulation resistance test shall be conducted in accordance with Method 302, test condition "A" of MIL-STD-202 and shall meet the specified requirements. Insulation resistance measurements shall be made between insulated points and ground.

4.6.10.2 Continuity. - The continuity of the wiring shall be tested using a conventional continuity tester to demonstrate continuity in accordance with the applicable wiring diagram.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with packaging Data Sheet 8587500. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Telescope Mount M138 is mounted on the top of the Quadrant Elevation M15 in 8 inch Howitzer M11O and 17mm Self-Propelled Gun M107 to mount an Elbow Telescope of the M116 Series for direct fire".

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of-this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable. stock number
(d) Provisions for first article testing

Custodian:

Army-MU

Preparing activity:

Army - MU

Project No. 1240-A160

Source: https://assist.dla.mil -- Downloaded: 202640-22T05:30Z
Check the source to verify that this is the current version before use.

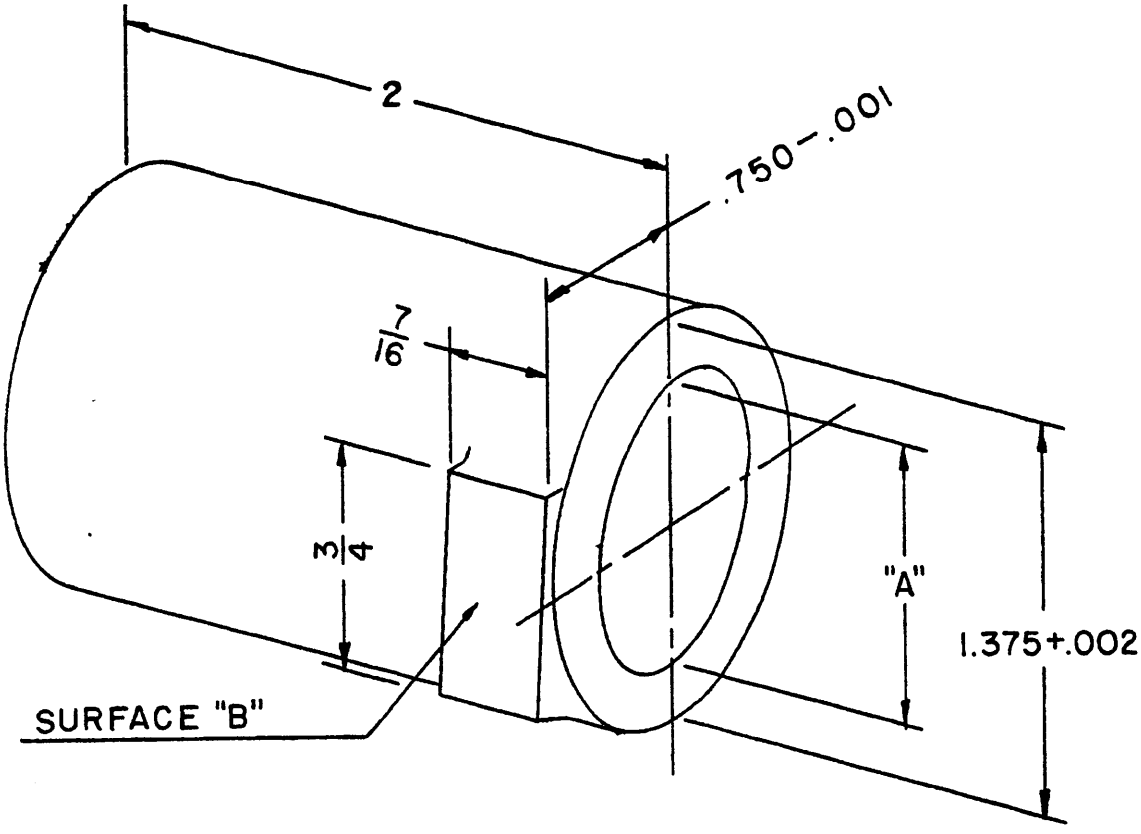


FIGURE I

NOTES: 1. Material brass-dimensions in inches.
2. Machine diameter “A” to securely hold auxiliary telescope.
3. Surface “B” to be parallel with vertical centerline of “A” within 0°3'.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:30Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T05:30Z
Check the source to verify that this is the current version before use.